

Generative AI as Seniority-Biased Technological Change: Evidence from U.S. Résumé and Job Posting Data*

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Abstract

This paper studies whether generative AI (GenAI) constitutes *seniority-biased technological change*, disproportionately reducing demand for junior workers. We develop a conceptual framework in which GenAI adoption reduces junior labor demand through task displacement and labor-saving productivity gains. We test the framework’s mechanisms and implications using U.S. résumé data covering 65 million workers at more than 280,000 firms, allowing us to track firm-level employment by seniority. GenAI adoption is identified through text analysis that detects ‘GenAI integrator’ job postings, signaling active GenAI implementation by firms. Following adoption, junior employment declines in adopting firms relative to non-adopters, while senior employment trends remain largely unchanged. This decline is concentrated in occupations most exposed to GenAI and is driven primarily by slower hiring rather than increased separations. Moreover, in adopting firms, GenAI-exposed tasks become increasingly less likely to appear in junior task bundles.

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ment positions, including job titles, start and end dates, and the employing firm.¹⁶

A key feature of the dataset is the standardized *seniority level* variable for each position, constructed by Revelio through an ensemble modeling approach based on multiple sources of information. This measure combines information from (i) the worker’s current job (title, firm, and industry), (ii) their work history (tenure and previous seniority), and (iii) their age. These three inputs produce separate scores, which are averaged and then categorized into seven standardized seniority levels: Entry Level, Junior Level, Associate Level, Manager Level, Director Level, Executive Level, and Senior Executive Level. In the analysis that follows, we group positions into two broad categories: *juniors* (Entry and Junior) and *seniors* (Associate and above).¹⁷ Supplemental Appendices A.1 and A.2 provide comprehensive data validation; Section 4.3 below summarizes their main findings.

We merge the worker résumé data with Revelio’s job postings database at the firm level. The job postings database tracks firms’ recruitment activity since September 2021 and contains, for each posting, a firm identifier, posting date, and the raw text of the job description. We use these job-posting texts to construct our measure of firm-level GenAI adoption, as described in Section 4.2.

In addition, we incorporate the occupational GenAI exposure measures from Eloundou et al. (2024), merged with our position-level data via O*NET SOC codes. We rely on the GPT-4–based beta exposure measure (as in, for example, Brynjolfsson et al., 2025a), and classify all the positions into three categories: low exposure (0–25th percentile), medium exposure (25–75th percentile), and high exposure (75–100th percentile).

We also construct a representative sample of *tasks* from Revelio’s job postings database. The extraction pipeline is motivated by Hampole et al. (2025), who construct firm-level AI exposure by mapping résumé text to O*NET tasks. From the Revelio job-postings corpus we draw a representative sample of 1,000 firms containing 355,013 postings, and map each posting’s free-text description to a subset of the 19,265 standardized O*NET tasks using a two-stage pipeline. The first stage is a TF-IDF retriever that returns the top 50 candidate O*NET tasks for each posting; the second stage is a large language model

¹⁶The initial data download was performed in May 2025. In April 2026, we re-downloaded the positions data to capture profile updates that users added with a lag, ensuring more complete coverage of recent employment spells.

¹⁷More details on Revelio’s seniority classification methodology are available at <https://www.data-dictionary.reveliolabs.com/methodology.html#seniority>.

tions and zero for low-exposure occupations; γ_{it} are firm-by-time fixed effects; and $\xi_{p(i)ct}$ are industry-by-exposure-by-time fixed effects. Equation 12 has the same structure as the seniority-based triple-difference in Equation 11, but here the third dimension is *occupational exposure* (high vs. low) rather than seniority. The coefficients β_j^s trace, separately for juniors and seniors, how employment in high-exposure occupations evolves relative to low-exposure occupations within the same firm and quarter, comparing adopters to non-adopters.

Figure 6 plots the estimated β_j^s coefficients for juniors and seniors, with both series normalized to zero in 2022Q4, the last pre-ChatGPT quarter. Two patterns stand out. First, throughout the pre-ChatGPT period, high-exposure occupations expand faster than low-exposure occupations at adopters relative to non-adopters for both seniority groups, consistent with adopters tilting their workforce toward exposed occupations over time. The two profiles track each other closely through 2022, with the senior coefficients rising somewhat faster than the junior coefficients. Second, beginning in 2023Q1, the two series diverge sharply. The junior series declines steadily, reaching a roughly 7 log point decline in high-exposure versus low-exposure junior employment at adopters by 2025Q1, relative to the 2022Q4 baseline. In contrast, the senior series continues to rise after 2022Q4, with no visible break around the ChatGPT launch.

Taken together, these results provide two insights. First, the post-2022 contraction in junior employment at adopters is concentrated in high-exposure occupations rather than reflecting a broad-based decline across all junior roles. Second, the design helps rule out a purely compositional explanation—namely, that high-exposure occupations declined economy-wide after 2023 and adopters simply employed a larger share of such roles. By comparing high- and low-exposure occupations *within* the same firm-by-quarter, and by running the same specification for seniors as a placebo-like comparison, we find that high-exposure occupations contract relative to low-exposure occupations only for juniors and only at adopters after the ChatGPT launch. The senior profile continues along its pre-existing upward trend, consistent with the interpretation that GenAI adoption is seniority-biased: it shifts demand away from junior workers in the very occupations where the technology is most applicable, while leaving senior employment in those same occupations essentially unaffected.

Supplemental Appendix A.3 reports a complementary heterogeneity exercise that re-

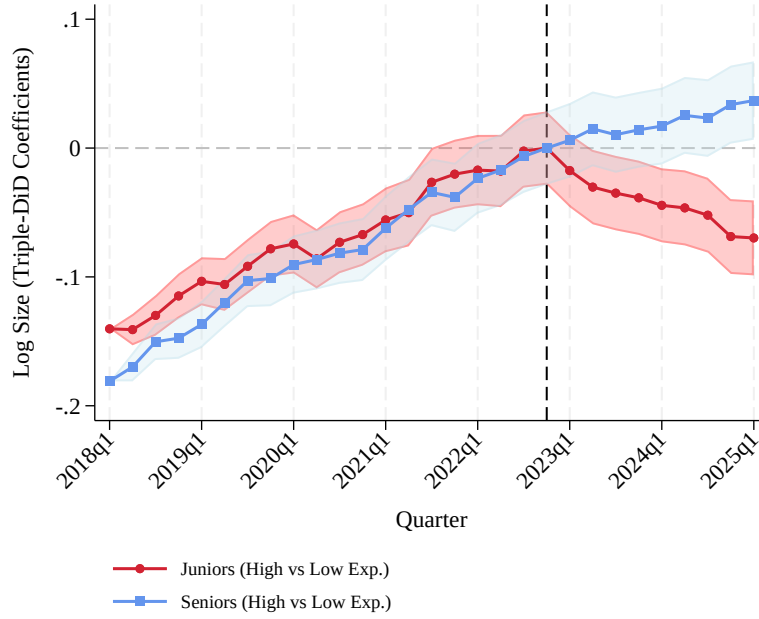


Figure 6: Triple-Difference by Occupational Exposure—Juniors vs. Seniors

Notes: This figure reports the estimated coefficients β_j^s from Equation 12, estimated separately for juniors (red) and seniors (blue). Each series captures the evolution of log employment in high-exposure occupations relative to low-exposure occupations at adopters relative to non-adopters, conditional on firm-by-time and industry-by-exposure-by-time fixed effects. Details on the exposure measure are provided in Section 4.1. Shaded bands are 95% confidence intervals. Standard errors are clustered at the firm level.

estimates the DiD specification within tiers of juniors' educational background, measured by the prestige of their alma mater. The relative decline in junior employment is U-shaped across tiers: largest for juniors from mid-tier institutions and smallest for juniors from the most and least selective institutions.

5.2.4 Discussion: The Timing of the Junior Employment Decline

The early and pronounced decline in the triple-difference coefficients—starting to emerge in 2023Q1, shortly after the release of ChatGPT—may appear surprising, given that automation effects typically unfold more gradually. One plausible explanation is that the rapid diffusion of GenAI tools at the end of 2022 constituted a discrete shock to firms' expectations about future automation. In response, firms may have adjusted in a forward-looking manner by scaling back hiring for junior roles they anticipated would be automated in the near future. Such preemptive adjustments could be optimal for firms if

Table 4: Hiring, Separations, and Promotions

	Hiring	Separation	Promotion (rate)
Treat $\times$ Post	−4.006*** (0.221)	−1.089*** (0.163)	0.033** (0.014)
Pre-period mean	5.059	4.154	0.822
Observations	7,941,179	7,941,179	7,690,581
Clusters (firms)		280,866	

Notes: This table reports the estimated β from Equation 15. Estimation begins in 2018 to exclude early-sample trends. Standard errors clustered by firm in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

This paper provides early, large-scale evidence that the diffusion of GenAI since 2023 is associated with *seniority-biased* employment effects within firms. Using résumé and job-posting data linked to roughly 281,000 U.S. firms, together with a firm-level adoption measure based on “GenAI integrator” vacancies, we document a decline in junior employment relative to senior employment at adopting firms compared with non-adopting firms. This decline is concentrated in high-exposure occupations: within adopting firms, junior employment in high-exposure occupations contracts relative to low-exposure occupations after 2022, while the corresponding senior series continues to rise. Task-level evidence from job postings points to the same mechanism: GenAI-exposed tasks become less likely to appear in junior task bundles at adopting firms. Finally, decomposing employment flows shows that the decline is driven primarily by slower hiring rather than increased separations.

These findings should be interpreted with caution. Adopting firms differ systematically in size, workforce composition, and industry. Although our research designs absorb many observable and unobservable differences across firms, sectors, and time, unobserved confounding factors may remain. We therefore examine four prominent alternative explanations: monetary tightening, post-pandemic hiring dynamics, the expansion of remote work, and labor-supply shocks. These tests support our interpretation, but they cannot rule out all alternative explanations. In addition, our adoption measure captures deliberate organizational integration of GenAI but may miss informal or “silent” adop-